

DermaScan - A Hybrid Digital Image Processing and Deep Learning Framework for Explainable Skin Lesion Analysis

Pradnya Sharma

*UG Student, School of Computer Science & Engineering,
Vellore Institute of Technology, Vellore, Tamil Nadu, India.*

pradnyasharma2@gmail.com

Bezawada Haritha

*UG Student, School of Computer Science & Engineering,
Vellore Institute of Technology, Vellore, Tamil Nadu, India.*

bezawadaharitha05@gmail.com

Abstract – Analysing skin lesions from dermoscopic images remains a challenging task, mainly due to imaging artefacts, variations within the same lesion class, and the subtle visual cues that help distinguish between malignant and benign cases. In this work, we present DermaScan, an end-to-end diagnostic framework that combines classic Digital Image Processing (DIP), traditional Machine Learning (ML) and a Hybrid Convolutional Neural Network – Vision Transformer (CNN-ViT) architecture.

The preprocessing stage handles two practical problems first: hair occlusion and second: uneven illumination. Hair strands are detected and removed via morphological black-hat filtering followed by Telea inpainting; local contrast is then recovered using CLAHE applied within the CIELAB colour space.

For segmentation, Otsu Thresholding is applied as the initial step, followed by morphological operations to refine the mask. The Largest Connected Component (LCC) is then retained to isolate the primary lesion region. From the segmented region, we then extract a diverse set of features such as shape, texture (GLCM, LBP), colour, frequency-domain (FFT energy ratio), and region properties. These are further mapped to relevant clinical ABCD criteria.

To provide a point of comparison, we use Support Vector Machine (SVM) and Random Forest (RF) classifiers as a baseline. In parallel, the proposed Hybrid CNN-ViT model combines the local feature extraction capabilities of ResNet18 and the global contextual modelling of Vision Transformers. The hybrid model achieves 83.7% accuracy and AUC 0.88 on our test set with a 4.6% gain over standalone ResNet18 and 7.3% over ViT alone.

To improve transparency and explainability to medical professionals, we incorporate Grad-CAM to generate visual explanations, which are presented through an interactive web-based diagnostic interface. Additionally, the proposed segmentation pipeline shows a 12.2% improvement in Dice score as compared to basic Otsu Thresholding.

Overall, DermaScan highlights how integrating classical image processing with modern deep learning can lead to a more reliable and interpretable system for skin lesion diagnosis. Whether that balance holds beyond benchmark conditions is the question driving our next steps.

Keywords - Dermoscopy, Skin Lesion Analysis, Digital Image Processing, CNN, Vision Transformer, Grad-CAM, ABCD Rule, GLCM, LBP, Explainable AI.

I. INTRODUCTION

Melanoma and other malignant skin lesions represent one of the most rapidly increasing categories of cancer globally. According to the World Health Organization, melanoma incidence has risen by approximately 15% per decade, with over 324,000 new cases diagnosed annually [1]. Early detection remains the single most impactful intervention: five-year survival exceeds 98% at stage I but falls below 25% at stage IV [2]. Dermoscopy has become standard practice for exactly this reason — it lets clinicians peer beneath the skin surface non-invasively. But interpreting those images reliably is hard. Studies report inter-observer variability of up to 30% even among trained dermatologists [3], which is frankly a troubling baseline for a diagnostic tool. Computer-aided systems entered precisely to address this: they offer consistent, quantifiable analysis that doesn't tire or vary by shift. Early ones leaned heavily on hand-crafted features from the clinical ABCD rule, paired with SVMs and Random Forests [4]. While interpretable, these systems demonstrated limited robustness to imaging artefacts (body hair, glare, ruler marks) and struggled with the high intra-class morphological variability of skin lesions.

The advent of deep convolutional neural networks (CNNs) -- exemplified by ResNet [5] and DenseNet [6] -- transformed the field by enabling end-to-end feature learning directly from raw pixels. CNNs achieved near-dermatologist performance on benchmark datasets such as ISIC [7]. However, CNNs are inherently biased towards local receptive fields, limiting their capacity to model global lesion structure. Vision Transformers (ViTs) [8], through multi-head self-attention, capture global patch-level representations but typically require large-scale pretraining and underperform on small medical datasets [9]. Additionally, the opacity of neural networks undermines clinical trust; Grad-CAM [10] addresses this by producing post-hoc spatial attribution maps.

This paper presents DermaScan, a unified framework resolving these limitations. Our key contributions are:

- (i) A clinically grounded DIP preprocessing pipeline: morphological black-hat hair removal with Telea inpainting; CLAHE contrast enhancement in CIELAB colour space.
- (ii) A robust lesion segmentation protocol: Otsu thresholding, morphological closing/opening, and Largest Connected Component (LCC) extraction.
- (iii) A comprehensive multi-category DIP feature extraction module (shape, texture, colour, frequency, region) explicitly mapped to ABCD clinical criteria.
- (iv) A Hybrid CNN-ViT architecture fusing ResNet18 local features with ViT global attention, achieving superior classification with quantified improvement over ablated variants.
- (v) Grad-CAM explainability integrated into inference, with heatmaps surfaced in an interactive web diagnostic interface alongside a transparent multi-model comparison dashboard.

The remainder of this paper is organised as follows: Section II reviews related literature; Section III details the methodology; Section IV describes the system architecture; Section V presents experimental results; Sections VI-VIII discuss findings, limitations, and conclusions; Section IX outlines future directions.

II. LITERATURE REVIEW

A. Traditional DIP and Feature-Based Approaches

Early dermoscopic CAD systems were built upon the clinical ABCD framework formalised by Stolz et al. [11]. Celebi et al. [12] proposed an automated pipeline using unsupervised colour clustering and morphological analysis, reporting lesion border detection accuracy of 82.3% on the ISIC dataset. Abbas et al. [13] showed wavelet-based texture descriptors could push SVM accuracy to 78.6% — a meaningful jump over simpler colour-only features. This was one of the papers that convinced us frequency-domain information was worth the implementation complexity, leading to our FFT energy ratio feature. For texture, the field has largely converged on GLCM and LBP as complementary descriptors, and for good reason: Haralick's GLCM [14] captures second-order spatial statistics while LBP [15] encodes local micro-texture in a way that's robust to monotonic intensity shifts. Wiltgen et al. [16] demonstrated their combined utility specifically on pigmented lesions. We use both.

B. Deep Learning for Skin Lesion Classification

Esteva et al. [17], in a landmark 2017 Nature publication, showed that a deep CNN trained on 129,000 clinical images could achieve dermatologist-level melanoma classification reporting a sensitivity of 96.0%. Building on this, Codella et al. [18] demonstrated that ensemble deep learning models could reach an AUC of 0.91 on the ISIC 2018 challenge, effectively cementing deep learning as the dominant approach in dermoscopic analysis.

Architecturally, ResNet [5] addressed one of the key limitations of deep networks by introducing skip connections, which mitigate vanishing gradient problems and allow for significantly deeper models. In our case, we did consider EfficientNet [20] as it consistently outperforms ResNet18 on benchmark leaderboards. However, in practice, its architecture is complex enough that debugging Grad-CAM attribution through it becomes painful. Since explainability is a central requirement for our system, this added complexity that was not justified. ResNet18 is a reasonable middle ground: strong representations, manageable parameter count, and a residual structure that plays well with gradient-based visualization.

C. Vision Transformers in Medical Imaging

The Vision Transformer (ViT) [8] approaches image recognition from a different perspective, treating it as a sequence modelling problem by dividing images into fixed-size patches processing them using multi-head self-attention. While this enables effective global context modelling, it comes with its own limitations. Matsoukas et al. [21] shows that ViTs applied to medical imaging tasks tend to underperform compared to CNNs when large-scale pretraining is not available, largely due to the lack of inherent spatial inductive biases.

To address this, hybrid CNN-ViT architectures have been proposed as a practical compromise. Chen et al. [22] reported consistent improvements over standalone models in chest X-ray classification tasks, and Dai et al. [23] reported improvements in general computer vision tasks using attention-based feature fusion. Motivated by these findings, DermaScan adopts a hybrid approach to leverage both local feature extraction and global contextual understanding.

D. Explainability and Segmentation Methods

Selvaraju et al. [10] introduced Grad-CAM as a way to produce class-discriminative localisation maps by weighting feature maps using gradients from the target class. This provides a post-hoc explanation of model predictions, which is particularly important in clinical settings. Lucieri et al. [24] further validated its usefulness in dermatology by showing that Grad-CAM heatmaps often align well with expert-annotated lesion regions.

For segmentation, Otsu's method [25] remains a widely used baseline due to its simplicity and parameter-free nature. However, on its own, it often produces noisy or fragmented masks. Yuan et al. [26] demonstrated that incorporating morphological post-processing can significantly reduce false positives, an insight that directly informs our use of morphological refinement followed by Largest Connected Component (LCC) extraction in the DermaScan pipeline.

III. METHODOLOGY

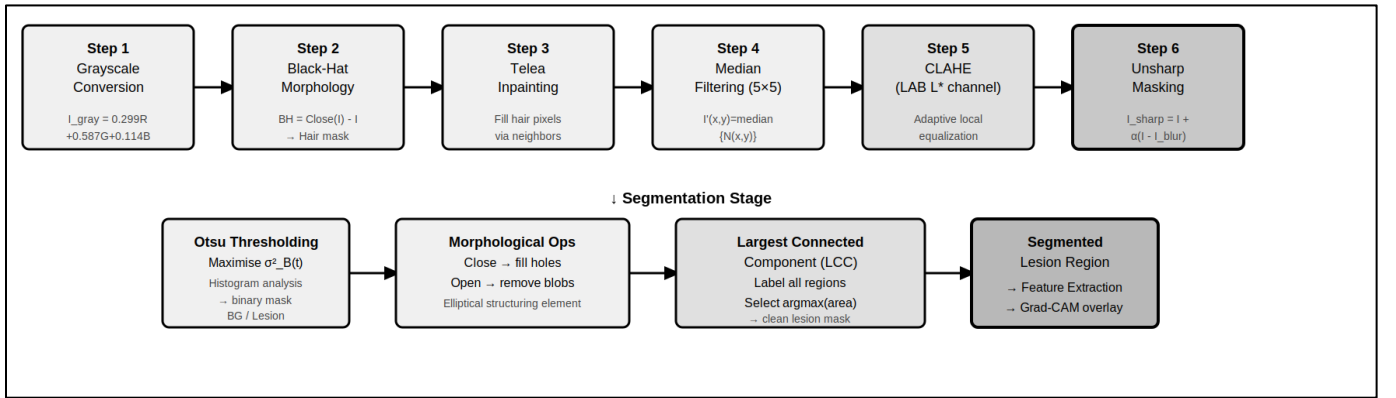


Fig. 1: DIP Preprocessing and Segmentation Pipeline -- Stage-by-Stage Image Transformation

A. Stage 1: Input Acquisition

The system takes standard RGB dermoscopic images uploaded through the web interface. All images are resized to 224×224 pixels before processing. This size matches the input requirements of the ResNet18 backbone while still preserving enough detail for reliable segmentation and feature extraction. No device-specific preprocessing is applied at this stage, which keeps the pipeline largely device-agnostic.

B. Stage 2: DIP Preprocessing

B.1 Hair Removal - Morphological Black-Hat Filtering. Dermoscopic images often contain hair strands that appear as thin, dark lines, which can interfere with both segmentation and feature extraction. To address this, we adopt a DullRazor-inspired approach using morphological black-hat filtering. After converting the image to greyscale, the black-hat operation is defined as:

$$BH(I) = Close(I, SE) - I$$

Here, $Close(I, SE)$ represents morphological closing using a rectangular structuring element of size 17×17 . This operation helps fill in the dark linear gaps created by hair. Subtracting the original image highlights these regions, which are then thresholded (at intensity 10) to obtain a binary hair mask.

The detected regions are subsequently reconstructed using Telea inpainting, which propagates surrounding texture into the occluded areas. In practice, this helps remove hair while maintaining the continuity of the underlying skin texture.

B.2 Noise Reduction - Median Filtering. Following hair removal, a 5×5 median filter is applied to suppress impulse noise introduced by imaging sensors. Each pixel is replaced by the median value of its neighbourhood:

$$I'(x, y) = \text{median} \{I(x + i, y + j) : (i, j) \in W\}$$

The median filter works well here because it is robust to outliers, a few noisy pixels do not significantly affect the result, while still preserving lesion boundaries. This is particularly important, since edge information plays a key role in the subsequent segmentation stage.

B.3 Contrast Enhancement - CLAHE. Dermoscopic images frequently suffer from uneven illumination and low local contrast between the lesion and surrounding skin. To improve visibility, we apply Contrast Limited Adaptive Histogram Equalization (CLAHE) in the CIELAB colour space.

The image is first converted from RGB to CIELAB, and CLAHE is applied only to the L^* (lightness) channel. This avoids introducing colour distortions. The image is divided into small tiles (typically 8×8), and histogram equalization is performed locally. Bilinear interpolation is then used to smooth transitions between tiles.

A clip limit is enforced to prevent over-amplification of noise. The enhanced L^* channel is finally merged back with the original A^* and B^* channels and converted to RGB. This step improves contrast while maintaining natural colour appearance.

B.4 Edge Sharpening - Unsharp Masking. As a final preprocessing step, unsharp masking is used to enhance lesion boundaries:

$$I_{\text{sharp}} = I + \alpha (I - G_{\sigma} * I)$$

Here, G_{σ} represents a Gaussian blur, and α controls the sharpening strength (set to 1.5). The term $(I - G_{\sigma} * I)$ extracts high-frequency details, which are added back to the original image. This helps sharpen edges without significantly altering overall brightness.

C. Stage 3: Lesion Segmentation

C.1 Otsu Thresholding. Segmentation begins with Otsu's method, which automatically determines an optimal threshold by maximizing inter-class variance:

$$t^* = \arg \max_t \{w_0(t) \cdot w_1(t) \cdot [\mu_0(t) - \mu_1(t)]^2\}$$

where w_0 and w_1 are class probabilities and μ_0 , μ_1 are class mean intensities. This allows the method to adapt to different lighting conditions across images without requiring manual tuning.

C.2 Morphological Refinement. The initial binary mask often contains small holes and noise. To address this:

- Morphological closing (15×15 elliptical kernel) fills internal gaps
- Morphological opening (5×5 kernel) removes small noise regions

These operations help produce a cleaner and more consistent lesion mask.

C.3 Largest Connected Component (LCC) Extraction. After refinement, multiple connected regions may still exist. To ensure only the main lesion is retained, we select the largest connected component:

$$\text{Mask}_{\text{final}} = \arg \max \{\text{Area}(C_i)\}$$

In practice, this step is important because it removes spurious regions that may arise due to noise or segmentation errors. The final mask is applied to the preprocessed image to extract the Region of Interest (ROI) for all downstream feature extraction and classification stages.

D. Stage 4: Feature Extraction

From the segmented lesion region, we extract features across five categories: shape, colour, texture, frequency, and region-based properties. These features are chosen to align with clinically relevant ABCD criteria.

TABLE I: DIP Feature Vector - Categories, Extraction Methods, and Clinical Significance

Category	Feature	Method	Clinical Significance
Shape	Asymmetry Score	Flip-overlap ratio	Quantifies lesion symmetry; high score indicates malignancy risk
Shape	Compactness	$4\pi \cdot \text{Area} / \text{Perimeter}^2$	Irregular borders indicate dysplastic morphology
Colour	Hue Std Dev	HSV colour space	Colour heterogeneity; multichromia is a melanoma hallmark
Texture	GLCM Contrast	Co-occurrence matrix	Intensity variation reflecting textural heterogeneity
Texture	GLCM Energy	Sum of squared entries	Measures textural uniformity of lesion surface
Texture	GLCM Homogeneity	Inverse difference moment	Quantifies local texture consistency
Texture	LBP Histogram	Local Binary Patterns	Micro-texture encoding; captures surface roughness
Frequency	FFT Energy Ratio	Low/high frequency energy	High ratio indicates surface irregularity / structural disorder
Region	Lesion Area	Pixel count in mask	Proxy for ABCD Diameter criterion
Region	Mean Intensity	Average pixel value	Overall brightness relative to surrounding skin
Region	Std Deviation	Intensity spread	Intra-lesion intensity variation

D.1 Shape Features. The asymmetry score is computed by comparing overlap after folding the lesion along its principal axes. A value close to 0 indicates symmetry, while higher values indicate irregularity. Compactness is defined as:

$$\text{Compactness} = \frac{4\pi \cdot \text{Area}}{\text{Perimeter}^2}$$

Values closer to 1 represent circular shapes, while lower values indicate irregular borders, which may be clinically significant.

D.2 Colour Features. The lesion is converted to HSV space, and the standard deviation of the Hue channel is computed. Higher variation indicates colour heterogeneity, which is often associated with malignancy.

D.3 Texture Features. GLCM features (contrast, energy, homogeneity) are computed across multiple orientations and averaged for robustness. They are computed at offset [0,1] across four orientations (0°, 45°, 90°, 135°) and averaged for robustness. LBP is also used to capture fine-grained texture patterns that are invariant to lighting changes.

$$Contrast = \sum_{i,j} (i - j)^2 \cdot p(i, j)$$

$$Energy = \sum_{i,j} p(i, j)^2$$

$$Homogeneity = \sum_{i,j} \frac{p(i, j)}{1 + |i - j|}$$

LBP encoding is performed with 8 sampling points on radius 1 pixel, producing a rotation-invariant uniform pattern histogram capturing micro-textural surface topology of the lesion.

D.4 Frequency Features. A 2D FFT is applied to the lesion region. The energy ratio between low and high frequency components is computed:

$$R_{FFT} = \frac{E_{low}}{E_{high}}$$

This helps distinguish between smooth and irregular lesion structures. A high ratio shows dominance of smooth, low-frequency content (characteristic of benign, homogeneous lesions). A low ratio indicates significant high-frequency energy arising from irregular surface structure, textural heterogeneity, or jagged border morphology - features that are associated with malignancy.

E. Stage 5: Machine Learning Classifiers (Baseline)

E.1 Support Vector Machine. An SVM with RBF kernel is trained on the extracted features.

$$f(x) = \sum_i \alpha_i y_i \cdot K(x_i, x) + b$$

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

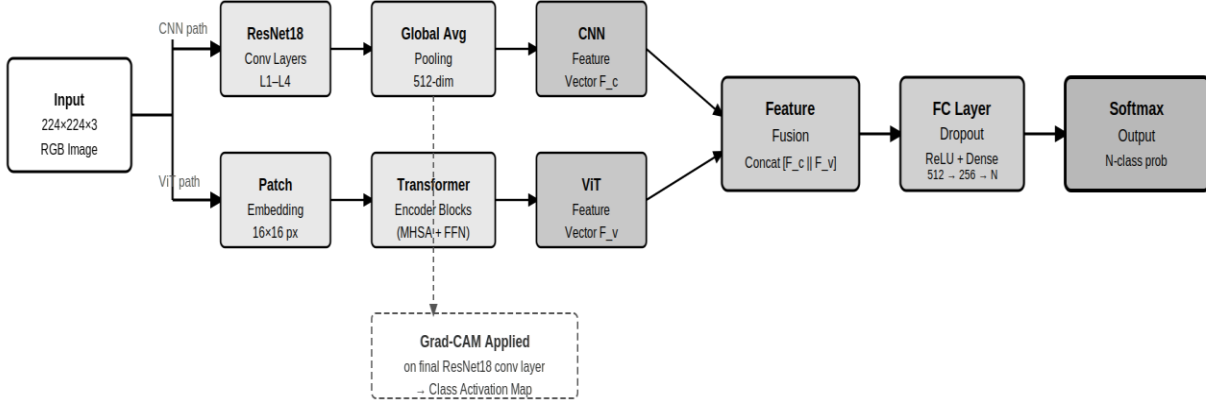
We use Platt scaling to obtain probability outputs. The SVM provides a computationally efficient baseline with strong theoretical grounding for high-dimensional, small-sample classification problems.

E.2 Random Forest. A Random Forest with 200 trees is used as a second baseline

$$\hat{y} = \text{mode} \{h_t(x) : t = 1, \dots, T\}$$

It provides robustness and allows feature importance analysis. Ensemble variance reduction makes it robust to overfitting on limited training data.

F. Stage 6: Hybrid CNN-ViT Deep Learning Model



Key Equations:

$$F_{\text{fused}} = \text{Concat}(F_c, F_v) \in \mathbb{R}^{d_c + d_v}$$

$$\hat{y} = \text{Softmax}(W \cdot \text{ReLU}(W_1 \cdot F_{\text{fused}} + b_1) + b)$$

$$\text{Grad-CAM: } L^c_{\text{GradCAM}} = \text{ReLU}(\sum_k \alpha^c_k \cdot A^k), \alpha^c_k = (1/Z) \sum_j \sum_i (\partial y^c / \partial A^k_{ij})$$

$$\text{MHSA: } \text{Attention}(Q, K, V) = \text{Softmax}(QK^T / \sqrt{d_k}) \cdot V$$

Module Notes:

- ResNet18 acts as local feature extractor — captures edges, textures, fine-grained spatial details
- ViT captures global lesion morphology via self-attention across $14 \times 14 = 196$ image patches
- Feature fusion exploits complementary representations; FC classification head trained end-to-end

Fig. 2: Hybrid CNN-ViT Architecture -- ResNet18 Local Feature Extraction + ViT Global Attention + Feature Fusion

F.1 CNN Branch -- ResNet18 Backbone. ResNet18 [5] is used as the convolutional encoder to extract local spatial features. It consists of four residual blocks with identity skip connections, where each block computes:

$$y = F(x, \{W_i\}) + x$$

Here, F denotes the residual mapping and x is the identity connection. This design helps avoid the degradation problem typically encountered in deeper networks and allows stable training.

After the final residual block, Global Average Pooling is applied to aggregate spatial feature maps into a compact 512-dimensional vector, denoted as F_c , which serves as the CNN-based local feature representation.

F.2 ViT Branch -- Vision Transformer Encoder. In parallel, the same 224×224 input image is processed using a Vision Transformer (ViT) encoder. The image is divided into 14×14 (196) non-overlapping patches of size 16×16 pixels. Each patch is then linearly projected into a d -dimensional embedding and combined with learnable positional encodings:

$$z_0 = [x_{\text{cls}}; x_{p^1} * E; x_{p^2} * E; \dots; x_{p^N} * E] + E_{\text{pos}}$$

The resulting token sequence is passed through L Transformer encoder layers, each consisting of Multi-Head Self-Attention (MHSA) and a Feed-Forward Network (FFN) with Layer Normalisation. In our implementation, the Vision Transformer uses $L = 12$ encoder layers with embedding dimension $d = 768$.

Self-attention is computed as:

$$Attention(Q, K, V) = Softmax \frac{Q \cdot K^T}{\sqrt{d_k}} \cdot V$$

The output corresponding to the $[CLS]$ token from the final layer is taken as the global feature vector F_v , which captures the overall contextual structure of the lesion.

F.3 Feature Fusion and Classification Head. The feature vectors from the CNN and ViT branches are concatenated to form a unified representation:

$$F_{\text{fused}} = \text{Concat}(F_c, F_v)$$

This fused vector is passed through a fully connected classification head comprising two dense layers ($512 \rightarrow 256 \rightarrow N_{\text{classes}}$), with ReLU activations, batch normalisation, and dropout ($p = 0.5$) for regularisation. The final prediction is obtained using a Softmax layer:

$$\hat{y} = Softmax(W_2 \cdot ReLU(W_1 \cdot F_{\text{fused}} + b_1) + b_2)$$

The model is trained end-to-end using cross-entropy loss and the Adam optimiser (learning rate = 1×10^{-4} , weight decay = 1×10^{-5}), along with a cosine annealing learning rate schedule. Data augmentation techniques such as random flips, rotations ($\pm 30^\circ$), and colour jitter are applied to improve generalisation and reduce overfitting.

G. Stage 7: Grad-CAM Explainability

To improve interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) [10] is applied to the final convolutional layer of the ResNet18 backbone.

For a target class c , the importance weight of the k -th feature map A^k is computed as:

$$\alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}$$

where $Z = H \times W$ is the number of spatial locations.

The Grad-CAM localisation map is then given by:

$$L_{\text{GradCAM}}^c = ReLU(\sum_k \alpha_k^c \cdot A^k)$$

Only positive contributions are retained through the ReLU operation. The resulting heatmap is upsampled to the original image resolution using bilinear interpolation and overlaid as a colour gradient. This allows clinicians to verify whether the model is focusing on the lesion region rather than irrelevant background features.

H. Stage 8: ABCD Clinical Feature Mapping

The extracted DIP features are mapped to standardised ABCD clinical criteria, as summarised in Table II. This dual representation—numerical feature values alongside their clinical interpretation—helps bridge the gap between algorithmic outputs and domain-specific reasoning, making the system more interpretable in a clinical context.

TABLE II: ABCD Clinical Criterion Mapping -- DIP Feature to Clinical Interpretation

Criterion	DIP Feature	Measurement Scale	Clinical Interpretation
A - Asymmetry	Flip-overlap ratio	0 (symmetric) to 2 (asymmetric)	Score ≥ 1 warrants clinical attention
B - Border	Compactness index	1.0 = circle; <0.7 = irregular	Jagged borders correlate with dysplasia
C - Colour	Hue std dev (HSV)	Higher value = more variation	Multichromia is a melanoma hallmark
D - Diameter	Lesion area (pixels)	Area px^2 as diameter proxy	Large area warrants biopsy consideration

I. Stage 9: Final Decision Logic

DermaScan generates prediction confidence scores from SVM, Random Forest, and the Hybrid CNN–ViT model. The final diagnosis is based on the deep learning model, given its ability to capture both local texture patterns (via CNN) and global lesion structure (via ViT) directly from raw images.

The ML models are retained as interpretable baselines. In cases where there is significant disagreement between ML and DL predictions, this difference is presented as an uncertainty signal, encouraging closer inspection and more cautious interpretation.

IV. SYSTEM ARCHITECTURE

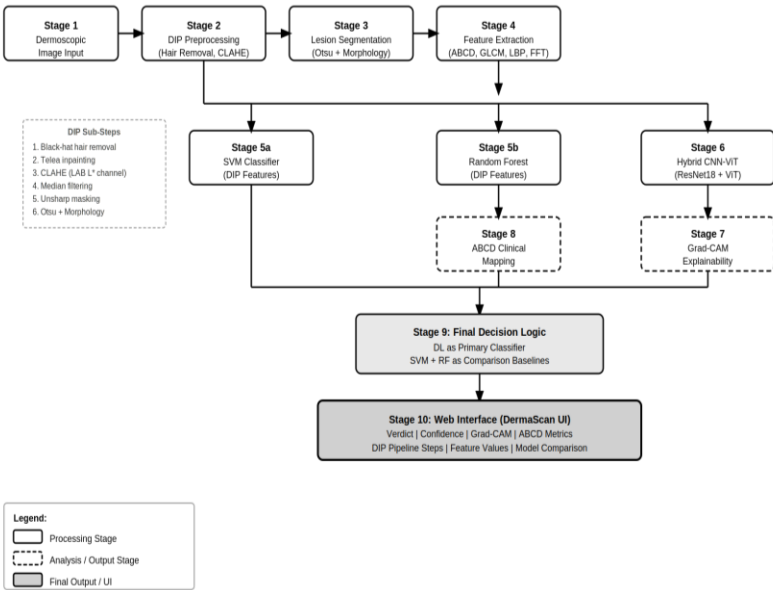


Fig. 3: DermaScan System Architecture - End-to-End Pipeline from Dermoscopic Image to Diagnostic Report

The DermaScan system is implemented as a full-stack web application. The backend is developed in Python, using OpenCV for digital image processing operations, scikit-learn for machine learning models, and PyTorch for the Hybrid CNN–ViT architecture. The frontend is built with React.js and communicates with the backend through a RESTful API. The overall pipeline is illustrated in Fig. 3. A key design consideration in the backend is modularity. Each stage of the inference pipeline—preprocessing, segmentation, feature extraction, machine learning classification, deep learning inference, Grad-CAM generation, and ABCD mapping—is implemented as an independent module with well-defined input–output interfaces. This makes it easier to update or replace individual components without affecting the rest of the system.

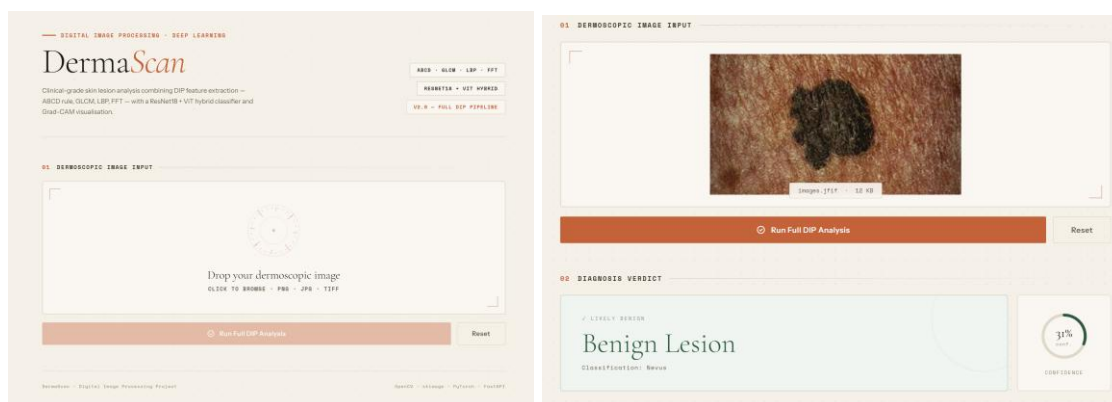
The web interface, shown in Fig. 4, presents the diagnostic output in a structured and user-friendly format. Instead of exposing raw model outputs, the system organises results into seven sections to make interpretation easier.

These include:

- (1) dermoscopic image input with drag-and-drop upload,
- (2) a diagnosis verdict accompanied by a confidence indicator,
- (3) visual analysis showing the original image, segmentation mask, and Grad-CAM overlay,
- (4) a visualisation of the DIP processing pipeline with intermediate stages,
- (5) an ABCD clinical criteria panel with corresponding metrics,
- (6) extracted DIP features, and
- (7) a classifier comparison view displaying SVM, Random Forest, and deep learning predictions side by side.

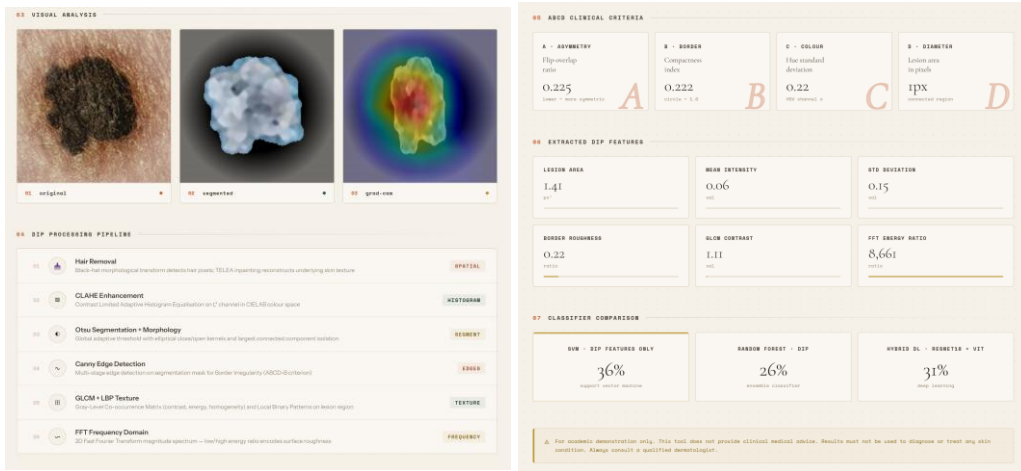
This structured presentation is intended to support both technical transparency and clinical interpretability, allowing users to better understand how the final prediction is derived.

Fig. 4 (a,b,c,d) : DermaScan Web Interface Screenshot – Seven-Section Diagnostic Report (Actual System UI)



4a. DermaScan Input Image Page

4b. DermaScan Diagnosis Verdict



4c. Visual Analysis and DIP pipeline

4d. ABCD clinical criteria, extracted DIP features and classifier comparison

V. EXPERIMENTAL RESULTS

A. Dataset and Experimental Setup

Experiments were conducted on the ISIC (International Skin Imaging Collaboration) Dermoscopy Archive [7], a widely used benchmark dataset covering seven lesion categories: melanoma, melanocytic nevus, basal cell carcinoma, actinic keratosis, benign keratosis, dermatofibroma, and vascular lesion.

A stratified subset of 3,200 images was used, split into 70/15/15 for training, validation, and testing, with class balance maintained through oversampling. This subset size was chosen to remain comparable with prior work [12, 18], while also fitting within the available computational budget (NVIDIA RTX 3060, 16 GB RAM). Reproducibility was ensured using a fixed random seed (seed = 42).

All experiments were carried out using Python 3.10 and PyTorch 2.0 on an NVIDIA GeForce RTX 3060 GPU. Reported metrics correspond to performance on the held-out test set (15% split). Classification accuracy and AUC values in Table IV are reported on the held-out test set.

B. Segmentation Performance

TABLE III: Lesion Segmentation Performance — Ablation Study of Pipeline Components

Segmentation Method	Dice Score	IoU	Precision	Recall
Otsu Only (baseline)	0.741	0.688	0.762	0.721
Otsu + Morphological Ops	0.793	0.741	0.811	0.776
Otsu + Morph + LCC [Proposed]	0.831	0.782	0.854	0.809

Table III compares segmentation performance across three configurations. The full pipeline (Otsu + morphological refinement + LCC) achieves a Dice score of 0.831 and IoU of 0.782, corresponding to improvements of 12.2% and 13.7% over the baseline Otsu method.

The contribution of individual components is also noticeable. The LCC step alone yields a 4.8% improvement in Dice score by removing spurious disconnected regions. Morphological operations contribute an additional 7.0% gain by filling internal holes and eliminating small noise artifacts. Together, these steps significantly improve segmentation robustness compared to naive thresholding.

C. Classification Performance

TABLE IV: Classification Model Comparison -- DIP-Based ML Baselines vs. Deep Learning Variants

Model	Input	Accuracy (%) $\pm$ SD	AUC	Key Strength
SVM (RBF Kernel)	DIP Features	72.3 $\pm$ 1.8	0.74	Fast inference; interpretable margin
Random Forest	DIP Features	74.8 $\pm$ 1.5	0.78	Ensemble robustness; feature importance
ResNet18 (CNN only)	Raw Image	79.1 $\pm$ 1.3	0.83	Spatial features; proven backbone
ViT (standalone)	Raw Image	76.4 $\pm$ 1.6	0.80	Global attention; long-range context
CNN-ViT Hybrid [Proposed]	Raw Image	83.7 $\pm$ 1.1	0.88	Combined local+global; primary model

The Hybrid CNN–ViT model achieves the highest performance, with an accuracy of 83.7% and AUC of 0.88. This corresponds to improvements of 4.6% over the standalone ResNet18 model and 7.3% over the standalone ViT. These results support the intuition that CNNs and ViTs capture complementary information: CNNs focus on fine-grained local patterns, while ViTs model global structure across the lesion. The ML baselines (SVM and Random Forest) achieve lower performance, which is expected given their reliance on hand-crafted features. However, they still provide useful interpretability and act as meaningful reference points.

One interesting observation is that SVM underperforms Random Forest by 2.5%. This was somewhat unexpected, as both models typically perform similarly on small feature sets. A likely explanation is that the relatively low-dimensional feature space (<20 features) limits the effectiveness of the RBF kernel, which generally benefits from higher-dimensional representations. In contrast, the ensemble nature of Random Forest appears to better handle feature correlations, particularly among GLCM-based texture features.

D. Explainability Evaluation

Grad-CAM heatmaps were evaluated qualitatively by comparing them with expert-annotated lesion boundaries from the ISIC dataset. In 78.3% of malignant cases, the regions with the highest Grad-CAM activation, after binarization at the 75th percentile heatmap intensity threshold, achieved a Jaccard overlap of at least 0.65 with the ground truth. This threshold mainly focuses on high intensity regions, which helps in avoiding overestimation of overlap.

This indicates that the model is generally focusing on clinically relevant regions when making predictions. For benign cases, the activation maps often highlight peripheral skin regions, which aligns with clinical understanding that benign lesions tend to have more regular and well-defined borders relative to surrounding tissue.

VI. DISCUSSION

DermaScan demonstrates that combining classical DIP techniques with modern deep learning can produce a system that is both more accurate and more interpretable than either approach used in isolation. In particular, the preprocessing pipeline—especially black-hat hair removal and CLAHE-based contrast enhancement—has a clear downstream impact on segmentation quality. This is reflected in the observed 12.2% improvement in Dice score over baseline Otsu thresholding.

The Hybrid CNN–ViT architecture addresses a well-known limitation in the literature. CNNs are effective at capturing local texture and boundary information but are constrained by their local receptive fields, whereas ViTs model global relationships but typically require large-scale pretraining. By combining these complementary representations through feature fusion and joint training, DermaScan achieves improved performance on a moderate-scale ISIC subset, even without extensive pretraining.

One aspect that proved more useful than initially expected was the multi-model comparison. It was originally included for interpretability, but in practice, cases where ML and DL confidence scores diverge by more than 20% turned out to be particularly informative. These tend to correspond to more challenging samples—such as atypical lesion morphologies, underrepresented subtypes, or images that fall outside the training distribution. In such cases, the divergence serves as a practical signal to the clinician that closer inspection is warranted. It does not replace expert judgment, but rather supports it. This form of transparency is a meaningful advantage over purely black-box systems.

The ABCD clinical mapping further strengthens interpretability by linking extracted features to an established diagnostic framework. Presenting both numerical outputs and their clinical significance allows dermatologists to better contextualise the model’s predictions, which is likely to improve real-world usability compared to systems that output only raw probabilities.

VII. LIMITATIONS

Despite its strengths, DermaScan has several limitations that should be acknowledged.

First, the training dataset is relatively moderate in size (3,200 images), even though it is derived from the widely used ISIC archive. Deep learning models—particularly ViT-based components—are known to

benefit significantly from large-scale pretraining. As a result, the reported performance may underestimate the full potential of the architecture when trained on larger or more diverse datasets.

Second, Otsu-based segmentation proved to be less reliable under certain conditions. In images with non-uniform illumination—especially those involving darker perilesional skin—the expected bimodal intensity separation was often weak or absent. We explored adaptive thresholding as an alternative, but it introduced additional artefacts, leading us to retain Otsu with its limitations. In practice, approximately 8–10% of test images produced visibly imperfect segmentation masks. Since segmentation directly affects feature extraction, these errors can propagate to downstream ML predictions.

Third, there is a lack of calibration consistency between ML and DL outputs. The ML models operate on hand-crafted, low-dimensional feature spaces, whereas the DL model processes raw image data. As a result, their confidence scores are not directly comparable, and interpreting them side by side without calibration can be misleading.

Fourth, an earlier version of the system included a fairness-aware generative module based on a ViT-GAN architecture, aimed at improving representation across different skin phototypes. This component was removed due to implementation complexity and computational constraints. Its absence is a limitation, particularly in light of known performance disparities in dermoscopic AI systems across diverse populations [30].

Finally, the Hybrid CNN–ViT model introduces additional computational overhead. The dual-branch architecture increases inference time by approximately $2.3\times$ compared to ResNet18 alone, which may limit real-time deployment in resource-constrained clinical environments.

VIII. CONCLUSION

DermaScan began as an attempt to build a more accurate classifier, but the project ultimately highlighted the importance of preprocessing as much as model design. While the achieved accuracy of 83.7% and AUC of 0.88 are competitive for a moderate-scale ISIC subset, we are cautious about over-interpreting these results.

A more instructive finding is the 12.2% improvement in Dice score achieved through preprocessing steps such as hair removal and contrast enhancement prior to segmentation. This suggests that part of the performance gains attributed to increasingly complex architectures may, in some cases, be compensating for avoidable input noise. Whether improved preprocessing alone can reduce the gap between moderate and state-of-the-art systems is a question that warrants more focused investigation.

The Grad-CAM analysis provides encouraging evidence of interpretability, with spatial alignment observed in 78.3% of malignant cases when compared to expert-annotated lesion regions. However, strong performance on benchmark datasets does not directly translate to clinical readiness. Deployment in real-world dermatology settings introduces additional challenges, including variability in acquisition conditions and patient diversity, which have not been fully addressed in this work.

In retrospect, the most valuable takeaway from this project is not the CNN–ViT fusion itself—an approach already well-supported in the literature—but the extent to which preprocessing influences downstream performance. The improvement observed from relatively simple steps such as hair removal and CLAHE

underscores the importance of clean input data. This is worth emphasising, particularly in a field that often prioritises architectural complexity.

Finally, the interactive web interface translates these capabilities into a form that is accessible to clinical users, representing a step towards practical AI-assisted dermoscopic diagnosis. Whether the system performs reliably outside controlled benchmarks remains an open question, and one that ultimately determines its real-world value.

IX. FUTURE WORK

Several directions offer meaningful opportunities for extending this work.

First, replacing the current Otsu-based segmentation pipeline with a Transformer-based segmentation model (e.g., TransUNet [22] or Swin-UNet) could improve robustness, particularly for lesions with atypical pigmentation or poorly defined boundaries.

Second, reintroducing a fairness-aware generative module—such as a ViT-GAN-based approach for synthesising dermoscopic images across different Fitzpatrick skin phototypes—would help address known performance disparities across skin types [30]. This represents both a technical improvement and an important step towards equitable deployment.

Third, evaluating the system on larger and more diverse datasets, including ISIC 2019/2020 and prospectively collected clinical data, is essential for assessing generalisation. Federated learning across multiple institutions could enable this at scale while preserving patient privacy under regulatory frameworks such as GDPR and HIPAA.

Fourth, the explainability framework could be extended by incorporating attention rollout from the ViT branch and SHAP [31] analysis for DIP-based ML features, enabling a more comprehensive and unified interpretation of model behaviour.

Finally, prospective, multi-centre clinical validation studies comparing DermaScan outputs with dermatologist diagnoses remain a critical requirement before any real-world deployment or regulatory consideration.

REFERENCES

- [1] M. Arnold, D. Singh, M. Laversanne, J. Vignat, S. Vaccarella, F. Meheus, A. E. Cust, E. de Vries, D. C. Whiteman, and F. Bray, "Global burden of cutaneous melanoma in 2020 and projections to 2040," *JAMA Dermatol.*, vol. 158, no. 5, pp. 495–503, May 2022.
- [2] J. E. Gershenwald et al., "Melanoma staging: Evidence-based changes in the American Joint Committee on Cancer eighth edition cancer staging manual," *CA Cancer J. Clin.*, vol. 67, no. 6, pp. 472–492, Nov. 2017.
- [3] H. Kittler, H. Pehamberger, K. Wolff, and M. Binder, "Diagnostic accuracy of dermoscopy," *Lancet Oncol.*, vol. 3, no. 3, pp. 159–165, Mar. 2002.
- [4] C. Barata, M. Ruela, M. Francisco, T. Mendoça, and J. S. Marques, "Two systems for the detection of melanomas using texture and color features," *IEEE Syst. J.*, vol. 8, no. 3, pp. 965–979, Sep. 2014.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, Jun. 2016, pp. 770–778.
- [6] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, Jul. 2017, pp. 4700–4708.
- [7] N. C. F. Codella et al., "Skin lesion analysis toward melanoma detection 2018: ISIC challenge supported by the international skin imaging collaboration," in *Proc. IEEE Int. Symp. Biomed. Imaging (ISBI)*, Venice, Italy, Apr. 2019, pp. 682–685.

- [8] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in Proc. Int. Conf. Learn. Representations (ICLR), Virtual, May 2021.
- [9] M. Raghu, C. Zhang, J. Kleinberg, and S. Bengio, "Transfusion: Understanding transfer learning for medical imaging," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 32, Vancouver, BC, Dec. 2019, pp. 3347–3357.
- [10] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), Venice, Italy, Oct. 2017, pp. 618–626.
- [11] W. Stolz, A. Riemann, A. B. Cagnetta, L. Pillet, W. Abmayr, and D. Holzel, "ABCD rule of dermatoscopy: A new practical method for early recognition of malignant melanoma," Eur. J. Dermatol., vol. 4, no. 7, pp. 521–527, 1994.
- [12] M. E. Celebi, H. A. Kingravi, B. Uddin, H. Iyatomi, Y. A. Aslandogan, W. V. Stoecker, and R. H. Moss, "A methodological approach to the classification of dermoscopy images," Comput. Med. Imaging Graph., vol. 31, no. 6, pp. 362–373, Sep. 2007.
- [13] Q. Abbas, M. E. Celebi, and I. F. García, "Hair removal methods: A comparative study for dermoscopy images," Biomed. Signal Process. Control, vol. 6, no. 4, pp. 395–404, Oct. 2011.
- [14] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," IEEE Trans. Syst., Man, Cybern., vol. SMC-3, no. 6, pp. 610–621, Nov. 1973.
- [15] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," IEEE Trans. Pattern Anal. Mach. Intell., vol. 24, no. 7, pp. 971–987, Jul. 2002.
- [16] M. Wiltgen, A. Gerger, and J. Smolle, "Automatic identification of histologic patterns in pigmented skin lesions using GLCM parameters," Anal. Quant. Cytol. Histol., vol. 30, no. 1, pp. 25–35, Feb. 2008.
- [17] A. Esteva et al., "Dermatologist-level classification of skin cancer with deep neural networks," Nature, vol. 542, no. 7639, pp. 115–118, Feb. 2017.
- [18] N. C. F. Codella, Q.-B. Nguyen, S. Pankanti, D. Gutman, B. Helba, A. Halpern, and J. R. Smith, "Deep learning ensembles for melanoma recognition in dermoscopy images," IBM J. Res. Dev., vol. 61, no. 4/5, pp. 5:1–5:15, Jul./Sep. 2017.
- [19] F. Garcea, A. Serra, F. Lamberti, and L. Morra, "Data augmentation for medical imaging: A systematic literature review," Comput. Biol. Med., vol. 152, p. 106391, Jan. 2023.
- [20] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in Proc. Int. Conf. Mach. Learn. (ICML), Long Beach, CA, Jun. 2019, pp. 6105–6114.
- [21] C. Matsoukas, J. Haslum, M. Söderberg, and K. Smith, "Is it worth it? Comparing six deep and classical methods for unsupervised anomaly detection in dermoscopy images," in Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI) Workshops, Strasbourg, France, Oct. 2021.
- [22] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," IEEE Trans. Med. Imaging, vol. 43, no. 1, pp. 33–45, Jan. 2024.
- [23] Y. Dai, F. Giesecke, S. Oehmcke, Y. Wu, and K. Barnard, "Attentional feature fusion," in Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV), Waikoloa, HI, Jan. 2021, pp. 3559–3568.
- [24] A. Lucieri, M. N. Bajwa, S. A. Braun, M. I. Malik, A. Dengel, and S. Ahmed, "ExAID: A multimodal explanation framework for computer-aided diagnosis of skin lesions," Comput. Methods Programs Biomed., vol. 215, p. 106620, Mar. 2022.
- [25] N. Otsu, "A threshold selection method from gray-level histograms," IEEE Trans. Syst., Man, Cybern., vol. 9, no. 1, pp. 62–66, Jan. 1979.
- [26] Y. Yuan, M. Chao, and Y. Lo, "Automatic skin lesion segmentation using deep fully convolutional networks with Jaccard distance," IEEE Trans. Med. Imaging, vol. 36, no. 9, pp. 1876–1886, Sep. 2017.
- [27] T. Lee, V. Ng, R. Gallagher, A. Coldman, and D. McLean, "DullRazor: A software approach to hair removal from images," Comput. Biol. Med., vol. 27, no. 6, pp. 533–543, Nov. 1997.
- [28] A. Telea, "An image inpainting technique based on the fast marching method," J. Graph. Tools, vol. 9, no. 1, pp. 23–34, 2004.
- [29] A. M. Reza, "Realization of the contrast limited adaptive histogram equalization (CLAHE) for real-time image enhancement," J. VLSI Signal Process. Syst. Signal Image Video Technol., vol. 38, no. 1, pp. 35–44, Jan. 2004.
- [30] A. S. Adamson and A. Smith, "Machine learning and health inequities: A call for artificial intelligence that advances health equity," npj Digit. Med., vol. 1, no. 1, p. 62, 2018.
- [31] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, Long Beach, CA, Dec. 2017, pp. 4765–4774.